

Foundations of Feature selection

[Feature selection]

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Quadratic programming feature selection mRMR is a typical example of an incremental greedy strategy for feature selection: once a feature has been selected, it cannot be deselected at a later stage. While mRMR could be optimized using floating search to reduce some features, it might also be reformulated as a global quadratic programming optimization problem as follows:

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Hilbert-Schmidt Independence Criterion Lasso based feature selection For high-dimensional and small sample data (e.g., dimensionality > 105 and the number of samples < 103), the Hilbert-Schmidt Independence Criterion Lasso (HSIC Lasso) is useful. HSIC Lasso optimization problem is given as

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Application of feature selection metaheuristics This is a survey of the application of feature selection metaheuristics lately used in the literature. This survey was realized by J. Hammon in her 2013 thesis.

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$$JMI(f_i) = \sum_{f_j \in S} (I(f_i; c) + I(f_i; c | f_j)) = \sum_{f_j \in S} [I(f_j; c) + I(f_i; c) - (I(f_i; f_j) - I(f_i; f_j | c))] \\ \{\displaystyle \begin{aligned} JMI(f_i) &= \sum_{f_j \in S} (I(f_i; c) + I(f_i; c | f_j)) \\ &= \sum_{f_j \in S} [I(f_j; c) + I(f_i; c) - (I(f_i; f_j) - I(f_i; f_j | c))] \end{aligned}$$

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Here, $\overline{r_{cf}}$ is the average value of all feature-classification correlations, and $\overline{r_{ff}}$ is the average value of all feature-feature correlations. The CFS criterion is defined as follows:

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where $Q_{ii} = I(f_i; c)$ and $Q_{ij} = (I(f_i; c | f_j) + I(f_j; c | f_i)) / 2, i \neq j$. An advantage of SPECCMI is that it can be solved simply via finding the dominant eigenvector of Q , thus is very scalable. SPECCMI also handles second-order feature interaction.

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$$R(S) = \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \quad \{\displaystyle R(S) = \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j)\}$$

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$$\text{HSIC Lasso: } \min_{\mathbf{x}} \|\mathbf{L}^{-1} - \sum_{k=1}^n \mathbf{x}_k \mathbf{K}^{-1}(k)\|_F^2 + \lambda \|\mathbf{x}\|_1, \text{ s.t. } \mathbf{x}_1, \dots, \mathbf{x}_n \geq 0, \\ \{\displaystyle \text{HSIC Lasso: } \min_{\mathbf{x}} \|\mathbf{L}^{-1} - \sum_{k=1}^n \mathbf{x}_k \mathbf{K}^{-1}(k)\|_F^2 + \lambda \|\mathbf{x}\|_1, \text{ s.t. } \mathbf{x}_1, \dots, \mathbf{x}_n \geq 0,\}$$

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Subset selection Subset selection evaluates a subset of features as a group for suitability. Subset selection algorithms can be broken up into wrappers, filters, and embedded methods. Wrappers use a

search algorithm to search through the space of possible features and evaluate each subset by running a model on the subset. Wrappers can be computationally expensive and have a risk of over fitting to the model. Filters are similar to wrappers in the search approach, but instead of evaluating against a model, a simpler filter is evaluated. Embedded techniques are embedded in, and specific to, a model. Many popular search approaches use greedy hill climbing, which iteratively evaluates a candidate subset of features, then modifies the subset and evaluates if the new subset is an improvement over the old. Evaluation of the subsets requires a scoring metric that grades a subset of features. Exhaustive search is generally impractical, so at some implementor (or operator) defined stopping point, the subset of features with the highest score discovered up to that point is selected as the satisfactory feature subset. The stopping criterion varies by algorithm; possible criteria include: a subset score exceeds a threshold, a program's maximum allowed run time has been surpassed, etc. Alternative search-based techniques are based on targeted projection pursuit which finds low-dimensional projections of the data that score highly: the features that have the largest projections in the lower-dimensional space are then selected. Search approaches include:

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$$mRMR = \max_{S \subseteq S} \left[\frac{1}{|S|} \sum_{f_i \in S} I(f_i; c) - \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \right] \quad \{\displaystyle \mathit{mRMR} = \max_{S \subseteq S} \left[\frac{1}{|S|} \sum_{f_i \in S} I(f_i; c) - \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \right]\}$$

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In machine learning, feature selection is the process of selecting a subset of relevant features (variables, predictors) for use in model construction. Feature selection techniques are used for several reasons:

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■ L_1 regularization techniques, such as sparse regression, LASSO, and ■ L_1 SVM Regularized trees, e.g. regularized random forest implemented in the RRF package Decision tree Memetic algorithm Random multinomial logit (RMNL) Auto-encoding networks with a bottleneck-layer Submodular feature selection Local learning based feature selection. Compared with traditional methods, it does not involve any heuristic search, can easily handle multi-class problems, and works for both linear and nonlinear problems. It is also supported by a strong theoretical foundation. Numeric experiments showed that the method can achieve a close-to-optimal solution even when data contains >1M irrelevant features. Recommender system based on feature selection. The feature selection methods are introduced into recommender system research.

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Embedded methods have been recently proposed that try to combine the advantages of both previous methods. A learning algorithm takes advantage of its own variable selection process and performs feature selection and classification simultaneously, such as the FRMT algorithm.

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Exhaustive Best first Simulated annealing Genetic algorithm Greedy forward selection Greedy backward elimination Particle swarm optimization Targeted projection pursuit Scatter search Variable neighborhood search Two popular filter metrics for classification problems are correlation and mutual information, although neither are true metrics or 'distance measures' in the mathematical sense, since they fail to obey the triangle inequality and thus do not compute any actual 'distance' – they should rather be regarded as 'scores'. These scores are computed between a candidate feature (or set of features) and the desired output category. There are, however, true metrics that are a simple function of the mutual information; see here. Other available filter metrics include: